Sport Injuries & Joint Reconstruction

The Big Picture

Sport injuries split into **acute** (sudden, one trauma) vs **chronic** (overuse/repetitive). All come from **direct blows, torsion, or shearing** hitting soft tissue (skin, tendon, ligament) or hard tissue (bone). Know the named injuries, the **RICE** rule, and when to operate vs rehab.

Acute vs Chronic

- **Acute** — strains, sprains, dislocations, contusions, fractures. Sudden symptoms.
- **Chronic (overuse)** — tennis/golfer's elbow, shin splints, bursitis, stress fractures. Gradual.

Acute Management = RICE

- **R**est — 2–3 days, immobilise.
- **I**ce — ↓ pain/spasm, vasoconstriction, less cell death.
- **C**ompression — ↓ swelling.
- **E**levation — drains fluid back to heart.

Chronic: painkillers, immobilisation, steroid injection, physio, surgery, rehab.

Shoulder Instability

- Most unstable joint — **~50% of all dislocations**.
- Unstable from major trauma / recurrent micro-trauma OR unbalanced muscle pull.
- **Traumatic anterior instability = commonest (>95%)**. Mechanism: abduction + external rotation + extension.
- Recurrent dislocation → **Bankart lesion** (anterior labrum) + **Hill-Sachs lesion** (humeral head dent).
- **>50yo** dislocation often = rotator cuff tear.
- Recurrent subluxation = "**dead arm syndrome**" (catching, numbness, weakness overhead).
- Treat: relocate (Kocher/Cunningham) → **Bankart repair** (arthroscopic or open).

Knee Injuries

- Poorly supported joint between the 2 longest bones.
- **ACL** — usually **non-contact** (deceleration, planted foot + rotation). "**Pop**", gives way, fast swelling (<1hr), unstable. MRI. Operative reconstruction for young/athletes; conservative for elderly/sedentary.
- **PCL** — needs severe trauma (dashboard, motorcycle); often with other ligaments.
- **Meniscus** ("little moon") — absorbs shock, distributes load. Twisting injury. **Clicking, catching, locking**, joint-line pain, effusion. Treat RICE/NSAIDs → arthroscopic repair/excision. Asymptomatic tears left alone.

Ankle (commonest sport injury — 40% of all)

- **Inversion sprain = 75%** ("rolled ankle"). Hits **ATFL** (first/weakest), CFL, PTFL.
- Grading: **G1 stretch** (RTA 1–2wk), **G2 partial tear** (2–4wk), **G3 complete tear** (8–10wk).
- **Shin splints (medial tibial stress syndrome)** — runner overuse; women higher risk of progressing to stress fracture.
- **Achilles rupture** — men 30–40, "weekend warrior"; risk: **fluoroquinolones, cortisone**. ¼ misdiagnosed as sprain. Rupture at **watershed area 4–6cm above insertion**. Non-op casting in equinus (higher re-rupture) vs open repair.

Stress Fractures

- Tiny crack from repetitive load (runners — tibia, hip, foot, heel).
- **Extrinsic** (technique, surface, footwear, too-rapid program) vs **intrinsic** (age/osteoporosis, BMI extremes, anatomy, female menstrual irregularity).
- X-ray normal early → periosteal reaction; **MRI / bone scan** confirm.
- Treat: rest, ice, walking aids; operative + internal fixation if **femoral neck displaced**.

Arthroscopy ("look within the joint")

- Rigid optical scope, keyhole. Diagnostic + therapeutic (cartilage smoothing, ligament reconstruction, loose-body removal, biopsy, fracture fixation).
- **Advantages** — small incision, low morbidity, less inflammation, lower cost.
- **Complications** — intra-articular damage (commonest), haemarthrosis, infection, thrombophlebitis, instrument breakage.

Arthroplasty (joint replacement)

- **Indications** — advanced **OA**, RA, post-traumatic arthritis, **AVN** failing conservative care.
- **Contraindications** — paralytic/Charcot joints, **infected joint**, paediatric population.

Walk-Away Points

- **RICE** for acute; rehab/surgery for chronic.
- Anterior shoulder instability (>95%) → **Bankart + Hill-Sachs**.
- ACL = non-contact "pop" + fast effusion; meniscus = locking/clicking.
- Inversion ankle sprain (75%) hits **ATFL**; grade by RTA time.
- Achilles rupture: weekend warrior, fluoroquinolone/cortisone, watershed zone.

Sport Injuries & Joint Reconstruction — Revision Table

Concept	Detail
Acute injury	Sudden, single trauma — sprain, strain, dislocation, fracture, contusion
Chronic/overuse	Gradual — tennis/golfer's elbow, shin splints, bursitis, stress fracture
Mechanisms	Direct blow, torsion, shearing
RICE	**R**est (2–3d), **I**ce, **C**ompression, **E**levation

Shoulder instability	Detail
Frequency	~50% of all dislocations; most unstable joint
Commonest type	**Traumatic anterior (>95%)**
Mechanism	Abduction + external rotation + extension
Lesions	**Bankart** (labrum) + **Hill-Sachs** (humeral head)
>50yo dislocation	Think rotator cuff tear
Recurrent	"Dead arm syndrome"
Treatment	Relocate (Kocher/Cunningham) → **Bankart repair**

Knee injury	Key features	Treatment
ACL	Non-contact, "pop", effusion <1hr, unstable	Reconstruction (young/athlete)
PCL	Severe trauma, dashboard, +other ligaments	—
Meniscus	Twisting; clicking/catching/**locking**, joint line	Arthroscopic repair/excision

Ankle sprain	Detail
Frequency	Commonest sport injury (40%)
Inversion = 75%	Hits **ATFL** > CFL > PTFL
G1 stretch	RTA 1–2 wk
G2 partial tear	RTA 2–4 wk
G3 complete tear	RTA 8–10 wk

Other	Detail
Shin splints	Runner overuse; women → stress fracture risk
Achilles rupture	Men 30–40, weekend warrior; **fluoroquinolone/cortisone**; watershed 4–6cm above insertion
Stress fracture	X-ray normal early → MRI/bone scan; fix if femoral neck displaced

Procedure	Indications	Notes
Arthroscopy	Dx + Rx (cartilage, ligament, loose body)	Small incision, low morbidity; commonest complication = intra-articular damage
Arthroplasty	Advanced OA, RA, post-traumatic, AVN	**CI: infection, Charcot, paediatric**

Bold = high-yield.